

Sampling with multiple states

The **State variation** selector at the top left of the **States** page select the type of multiple states to use from **External digital**, **Static outputs** and **Dynamic outputs**. The controls shown in the dialog change dramatically as each type of multiple states is selected, so the three forms are described separately below.

What type of states do I need? For almost all purposes **Dynamic outputs** states are the most useful as this allows you to set different output pulses for each state. **External digital** states can be used only with external hardware that generates digital codes to indicate what is happening, while **Static outputs** states are limited to unchanging digital and DAC outputs, so both of these are more limited. The main part of this chapter will cover **Dynamic outputs** states with the other forms of multiple states described afterwards.

Dynamic outputs states

Dynamic outputs states are the most useful and general form of multiple states; each state generates a separate set of output pulses. The actual digital and DAC outputs for each state (along with some other information including the state label) are set by using the **Configure Pulses** dialog available from the **Outputs** tab of the sampling configuration, the states page is purely concerned with defining how many states there are and how they are sequenced – how **Signal** will switch from state to state during sampling.

The **Number of extra states** item controls the number of extra states, from 1 to 256. Note that this item sets the number of extra states in addition to state zero. Thus in the example shown below there are 5 states possible with codes running from 0 to 4. In many circumstances **Signal** will use only states 1 to 4, state zero is treated as an background or idle state. This also controls the states available in the pulses configuration dialog; you should set the extra states you want here before you set up the pulse outputs themselves.

State sequencing

Because in **Dynamic outputs** mode **Signal** is controlling the states (as it is in **Static outputs** mode), we need ways of defining which state is used when. The simplest way to do this is to control the state manually (which can be done using a control bar), but often we want some form of automatic states sequencing. **Signal** provides essentially two separate forms of states sequencing – numeric or protocol. In the numeric modes (there are three of these) simple numeric or randomised sequencing is provided while, in protocol mode a precisely defined complex sequence of states can be generated.